

Fragmented Housing Regulation as a Fiscal-Federalism Breakdown

The Distortion from Local Free-Riding, and the Leakage of State Preemption

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Abstract

Local housing regulation is provided by jurisdictions far finer than the labor market and housing submarkets they sit in. Each jurisdiction enjoys the local amenity of low density but does not internalize the regional cost of the housing supply it suppresses — the classic condition (??) under which decentralized provision of a local public good with interjurisdictional spillovers ceases to be efficient. This note frames fragmented housing regulation as exactly that breakdown and builds a small equilibrium model of the *market for regulation* aimed at two questions a purely descriptive account cannot answer. **First, how large is the distortion** from decentralized, non-internalized regulation, relative to a regional (internalized) benchmark? **Second, what is the leakage of state preemption** — when the state forces open one regulatory margin (the by-right zoning envelope), to what extent do jurisdictions reassert restriction through the discretionary margin it leaves untouched? The model's central empirical object is the cross-jurisdiction *reaction function*: in an efficient, no-spillover world jurisdictions would not respond to one another's regulation, so a nonzero strategic-interaction slope is the direct signature of the spillover that breaks the Decentralization Theorem. Every primitive maps to an item-level dataset built from planning-commission minutes paired with the by-right envelope, across Bay Area jurisdictions.

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1 The question, and why it is a federalism question

A large literature establishes that local land-use regulation suppresses housing supply in high-demand metros at large aggregate cost [???]. Recent work measures the *by-right* regulatory environment at scale, including with LLMs [??]. What that work measures is a level (or a set of levels) of stringency; what it largely takes as given is *why* regulation is set where it is, and what would happen under a different *assignment of regulatory authority*. The “homevoter” and “jurisdictional fragmentation” hypotheses supply the political ingredients [????], but not a tractable equilibrium object that prices the externality and supports counterfactuals.

The federalism frame. The Decentralization Theorem [?] holds that local provision of a local public good is efficient *absent interjurisdictional spillovers and scale economies*; ? recast the centralization trade-off as a political-economy problem, where provision is chosen by elected local representatives (decentralization) or a legislature (centralization), and the comparison turns on spillovers and taste heterogeneity operating through political channels. Housing regulation is precisely the case the theorem’s condition excludes. A jurisdiction that restricts development consumes a *local amenity* — low density, preserved “character,” a scarcity rent on incumbent housing — while the cost, suppressed housing supply, is borne *regionally*: by would-be entrants priced out of the metro, by the labor market that loses agglomeration when workers cannot locate near high-productivity jobs, and by every other jurisdiction in the same housing submarket whose prices the restriction props up. The benefit is internalized; the cost spills across boundaries onto people who do not vote in the restricting jurisdiction. Decentralized provision is therefore inefficient, and the inefficiency is not incidental — it is the structural reason fragmented housing regulation over-restricts.

What this buys, and the two questions. Framing the distortion as a federalism breakdown gives it a *named efficiency benchmark* (the regional, internalized optimum) and turns two otherwise-vague goals into sharp questions.

1. **The magnitude of the distortion.** How much regulation, and how much suppressed housing supply, is attributable to fragmentation per se — the gap between regulation chosen by atomistic localities each ignoring the spillover, and regulation chosen at the level that internalizes it (the submarket or the region)? This is the Oates wedge, made quantitative.
2. **The leakage of preemption.** State preemption is *partial centralization*: it forces open one regulatory margin and leaves another decentralized. If a jurisdiction regulates on two substitutable margins — a *by-right envelope* (zoning, the margin preemption targets) and a *discretionary friction* (planning-commission review, which it retains) — then preemption that loosens the envelope can be partly or wholly offset by tightening discretion. The leakage is the fraction of the intended loosening that localities claw back through the untouched margin. A descriptive measure of either margin alone cannot see it.

What is new here is the combination: a two-margin structure that makes preemption leakage a coherent object, embedded in a fragmented system whose strategic interaction is the empirical signature of the very spillover that makes decentralization inefficient.

2 Environment

One metro — the Bay Area — comprises J jurisdictions $j = 1, \dots, J$, grouped into $S \ll J$ housing submarkets $s(j)$, all drawing on *one* regional labor market. A worker employed anywhere in the metro earns the regional wage $w_m = w(N)$, $w' > 0$, $N \equiv \sum_j N_j$, capturing agglomeration rents [?]. This is the coupling that makes regulation’s cost regional: a household’s income depends on the metro, not on the jurisdiction it sleeps in, so the benefit of access is regional while the disamenity of the housing that provides it is local.

Preferences and prices. A household of type θ in j has $U = c + \alpha \ln h - \gamma g(D_j)$, $g' > 0$, where $-\gamma g(D_j)$ is the *local* disamenity of density (so low density is a normal good and high- θ residents are the natural demanders of restriction), and income $y(\theta) = w_m \theta$. Within submarket s , clearing gives the price index

$$p_s = \frac{\alpha N_s}{H_s}, \quad N_s = \sum_{j:s(j)=s} N_j, \quad H_s = \sum_{j:s(j)=s} H_j, \quad (1)$$

so housing built in any jurisdiction of s lowers the price faced by incumbents in *every* jurisdiction of s ; density $D_j = N_j/\bar{L}_j$ is local. (We treat h as a normalized unit of housing services so that H_s counts dwellings consistent with the data; the divisible-services reading and the indivisible-dwelling/spatial-indifference reading coincide under that normalization, and nothing below depends on the distinction.) Equations (1) contain the externality: the *cost* of approval (density) is local, the *benefit* (lower submarket price, more regional employment hence higher w_m) is shared.

3 Regulation on two margins

The empirical premise, visible in the minutes: among projects that reach a vote, approval (or its discretionary-review equivalent) is nearly universal. Restriction does not operate through denial at the hearing; it operates before and around it, on two margins.

The by-right envelope $\bar{z}_j$. The zoning map assigns each parcel a use and height/bulk district, defining the scale/use a project may obtain *ministerially* — with certainty, no hearing. Projects within the envelope never appear on the commission calendar; this is the margin LLM-based zoning measurement captures [?], and the margin state preemption targets.

The discretionary friction τ_j . A project exceeding the envelope (a use not principally permitted, a scale above the limit, a special-district trigger) enters discretionary review and is built only after a *discretionary tax* τ_j whose components the minutes record: process exposure (the share of activity forced off the by-right path, via `request.type`), mobilized opposition ($g(D_j)$ made operative, via stance-marked speaker counts), conditioning (`action=approved-with-conditions`), delay (continuances, via a case-level join), and rare denial. This is the margin LLM zoning measurement *cannot* see and the margin preemption leaves untouched.

Definition 1 (Effective regulation). *Effective stringency is $r_j = g(\bar{z}_j, \tau_j)$, increasing in the tightness of the envelope (smaller $\bar{z}_j$ pushes more activity into discretion) and in the discretionary tax. A jurisdiction restricts by lowering $\bar{z}_j$ or raising τ_j ; the two are substitutes in producing restriction but face different costs, and — crucially — state preemption acts on $\bar{z}_j$ but not τ_j .*

4 The homevoter's choice and the fragmentation externality

Each jurisdiction's regulation is chosen by a vote-maximizing authority answering to incumbent homeowners (share λ_j , weight ϕ_j), following the homevoter logic [?]. Mapping the two margins to two bodies resolves the unitary-authority shortcut: the council sets the slow map $\bar{z}_j$, the planning commission sets the fast discretionary tax τ_j . An incumbent's payoff, taking neighbors' policies as given, is

$$W_j^{\text{own}} = \underbrace{p_{s(j)}(\bar{z}_{-j}, \tau_{-j}; \bar{z}_j, \tau_j)}_{\text{submarket asset value}} - \underbrace{\gamma g(D_j(\bar{z}_j, \tau_j))}_{\text{local disamenity}}, \quad (2)$$

where $-j$ denotes the profile of all other jurisdictions (only co-submarket neighbors, $s(k) = s(j)$, enter with nonzero weight). The authority maximizes $\Omega_j = \phi_j \lambda_j W_j^{\text{own}} + (1 - \phi_j \lambda_j) W_j^{\text{ent}}$ over $(\bar{z}_j, \tau_j)$.

The free-riding wedge. The effect of j 's own approvals on the price its homeowners care about is attenuated by submarket size: $\partial p_{s(j)} / \partial (\text{supply}_j) = -(p_s / H_s)$. A jurisdiction small within its submarket bears the full *local* disamenity of any approval but captures little of the price relief, most of which leaks to co-submarket neighbors. So it under-supplies and free-rides on their construction — the microfoundation of the fragmentation externality.

Proposition 1 (Fragmentation over-restricts; the distortion is the Oates wedge). *Holding $(\phi_j \lambda_j, \gamma, \bar{L}_j)$ fixed, each jurisdiction's privately optimal restriction is increasing in the number of jurisdictions sharing its submarket: as j becomes atomistic the price-relief motive vanishes, the local-disamenity motive dominates, and both $\bar{z}_j$ tightens and τ_j rises. The **distortion** is the gap between the decentralized Nash regulation $\{(\bar{z}_j^*, \tau_j^*)\}$ and the regulation chosen by an authority that internalizes the submarket (or region) — the regional planner who values entrants' access to w_m and the cross-jurisdiction price externality in (1). This gap is the model's answer to Question 1, and it is exactly the inefficiency the Decentralization Theorem's spillover condition rules in.*

5 Reaction functions: the empirical signature of the spillover

Because $p_{s(j)}$ depends on neighbors' regulation, (2) is a game, with a reaction function $R_j(\bar{z}_{-j}, \tau_{-j})$ on each margin.

Proposition 2 (Regulation is strategically substitutable within a submarket). *For co-submarket jurisdictions, if a neighbor loosens (raises $\bar{z}_k$ or lowers τ_k), the submarket price falls, raising a homevoter-pivotal j 's incentive to restrict: j tightens. Neighbors' regulation are strategic substitutes; each free-rides on the others' supply, and the Nash equilibrium has less regional construction than the internalized optimum.*

Why this is the centerpiece. The reaction function is not merely an empirical regularity — it is the *direct test of the federalism breakdown*. Under the Decentralization Theorem's efficient case (no spillovers), jurisdictions would not respond to one another's regulation at all: each would serve its own residents independently and the cross-jurisdiction slope would be zero. A *nonzero* strategic-interaction slope is therefore the spillover, made visible — the empirical object whose sign and magnitude establish that the Oates condition fails and quantify how strongly. This is why estimating R_j is the core empirical task, and why a multi-jurisdiction panel (not a single metro) is required: single-jurisdiction data cannot, even in principle, reveal a reaction function.

6 Preemption and its leakage

State preemption (e.g. ministerial-approval mandates, by-right upzoning, the builder’s remedy) raises the marginal cost to j of a tight *envelope* — forcing $\bar{z}_j$ up or converting discretionary categories to ministerial ones — while leaving the marginal cost of the *discretionary* tax τ_j unchanged.

Proposition 3 (Leakage: substitution from envelope to discretion). *With effective restriction $r_j = g(\bar{z}_j, \tau_j)$ and the two margins substitutes in producing it, a preemption that raises $\bar{z}_j$ (the map loosens, as intended) induces a homevoter-pivotal jurisdiction that still demands restriction to raise τ_j in compensation (more items conditioned, more opposition entertained, more continuances). Effective restriction falls by less than the change in $\bar{z}_j$ alone implies, and **the conditional approval rate need not move**. The leakage is the share of intended loosening offset by the discretionary margin: 0 (preemption fully effective) to 1 (fully clawed back). This is the model’s answer to Question 2.*

Leakage is precisely what makes preemption a federalism question rather than a mechanical policy: partial centralization fails not because centralization is wrong but because it is *incomplete*, leaving a decentralized margin through which the spillover reasserts itself. Strategic substitutability (Prop. 2) compounds it: as preempted jurisdictions claw back via τ , co-submarket neighbors face a smaller price effect and adjust in turn.

7 Counterfactuals and the identification challenge

Both questions are counterfactual. Question 1 compares the observed decentralized equilibrium to an internalized benchmark never observed; Question 2 asks how supply responds to a change in the assignment of regulatory authority. Answering them requires recovering the structural primitives — the homevoter weights $\phi_j \lambda_j$, the disamenity $g(\cdot)$, the demand and agglomeration parameters — from *equilibrium* behavior alone, since only the chosen regulation is observed, not the objectives that generated it. This is the identification crux: the reaction-function slope is consistent with several mechanisms (the spillover modeled here, but also yardstick competition or common shocks; cf. the strategic-interaction literature ??), and disentangling them is what licenses the counterfactual. The empirical leverage comes from the *preemption ramp itself*: the staggered, rule-driven timing of state mandates (builder’s-remedy exposure following Housing-Element non-compliance) is a supply-side shock plausibly orthogonal to a jurisdiction’s contemporaneous demand shock, providing variation that moves $\bar{z}_j$ for reasons other than local taste — the wedge needed to separate the spillover from its observational rivals and to identify the leakage of Prop. 3.

8 Mapping to data

The by-right envelope $\bar{z}_j$ is read from zoning (and is largely available from existing LLM-based measurement, so need not be rebuilt); the discretionary tax τ_j is built fresh from planning-commission minutes, item by item, across Bay Area jurisdictions; the preemption ramp is dated from HCD Housing-Element compliance records; prices and quantities come from transaction data. The reaction-function estimation (Prop. 2) is a spatial-lag regression of a jurisdiction’s discretionary tax on co-submarket neighbors’, instrumented using the preemption variation of §7; the leakage estimation (Prop. 3) is a difference-in-differences around preemption exposure with the *envelope* and the *discretionary tax* as separate outcomes — predicting the map loosens while discretion tightens and the approval rate is flat. A short panel straddling the 2017–2023 preemption ramp suffices for both; the data infrastructure is documented separately.

9 Scope and a noted extension

The questions above are *static/short-panel* counterfactuals: the magnitude of the fragmentation distortion and the leakage of preemption are identified off the cross-section of jurisdictions and the preemption ramp, and do not require deep historical data. A natural but *separate* extension treats regulation as state-dependent — a “regulatory ratchet” in which restriction is path-dependent and preemption shocks have persistent or hysteretic effects, propagating across co-submarket neighbors over time. That dynamic question would require a long panel and confronts a harder identification problem (state dependence vs. unobserved heterogeneity); it is deliberately held as an extension rather than a core result, since the two headline questions do not depend on it.

10 Conclusion

Fragmented housing regulation is a textbook fiscal-federalism breakdown: a local public good (low density) whose provision via regulation imposes regional spillovers, so decentralized choice over-restricts. Two moves make the breakdown quantitative and policy-relevant — splitting regulation into a by-right envelope and a discretionary friction (so that preemption *leakage* is a coherent object), and embedding many jurisdictions in a game whose reaction functions are the direct empirical signature of the spillover. The model answers two questions a descriptive account cannot: how large the fragmentation distortion is, and how much of state preemption localities claw back through the margin it leaves untouched.